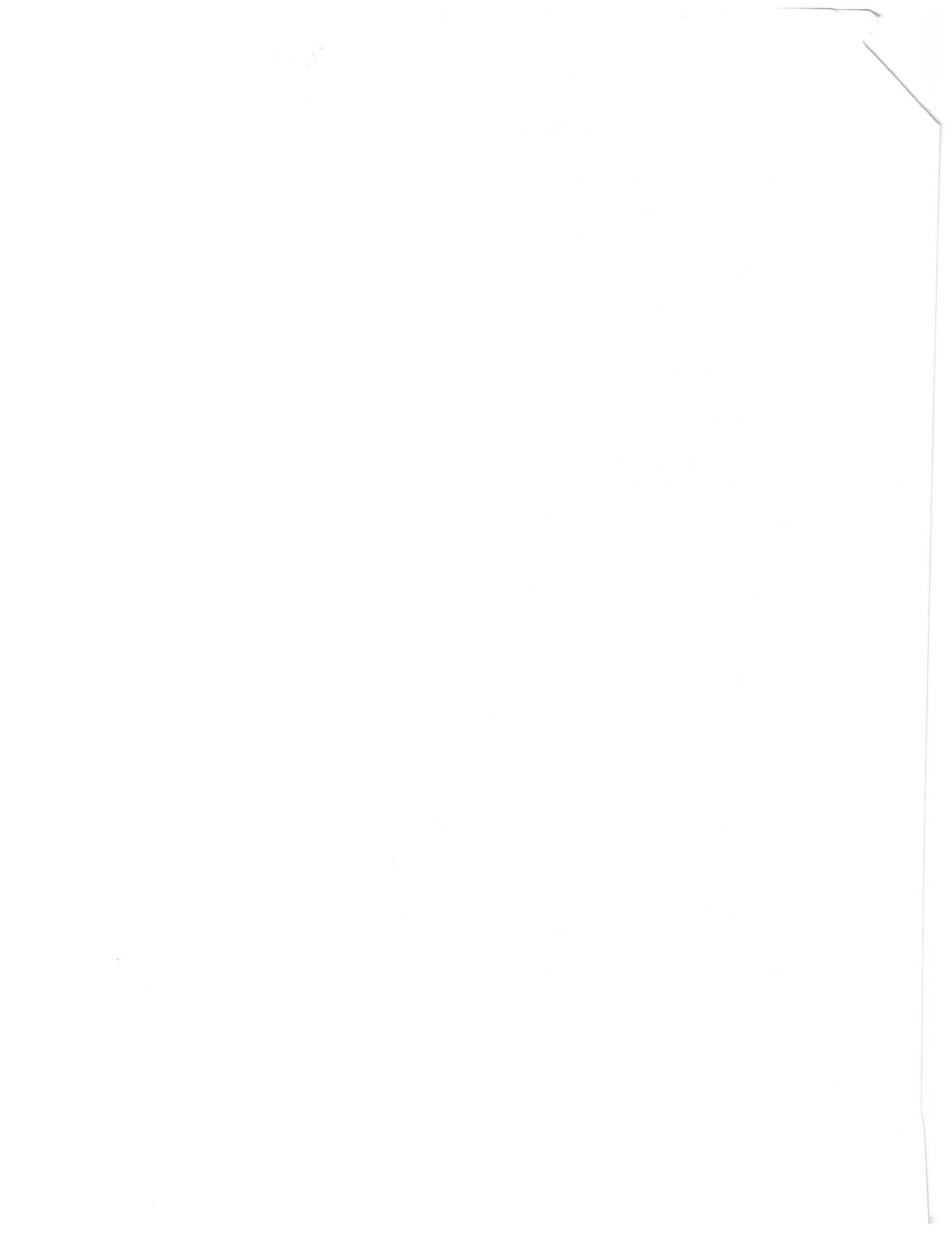


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Exam 4: NE533: Nuclear Fuel Performance
Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm .
 - a) Estimate the oxide thickness after this time?
 - b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume $\text{PBR} = 1.56$, $\rho_{\text{Zr}} = 6.5 \text{ g/cc}$, $\rho_{\text{ZrO}_2} = 5.68 \text{ g/cc}$.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.



Question 1.)

$$\bar{T} = 625^\circ\text{K} \quad t = 500 \text{ days} \quad t_0 = 600 \mu\text{m}$$

$$\text{Check if } t > t^* ; \quad t^* = 6.62 \cdot 10^{-7} \exp\left(\frac{11949}{625}\right) = 133.007 \text{ dgs}$$

$$t > t^*$$

$$\text{Find } \delta^* = 5.1 \cdot \exp\left(\frac{-550}{625}\right) = 2.1154 \text{ micron}$$

$$\text{find } k_L = 7.48 \cdot 10^6 \exp\left(\frac{-12500}{625}\right) = 0.015417$$

$$\delta = \delta^* + k_L(t - t^*) = 2.1154 + (0.015417)(500 - 133.007) = 5.6579 = \boxed{7.77 \mu\text{m}}$$

18/18

Part 2

$$f = 0.14 \quad P_m = 6.5 \quad P_{ox} = 5.68 \quad P_{BIR} = 1.56 \quad \frac{M_H}{M_o} = \frac{1}{16} = 0.0625$$

$$\int_0^1 z_{rO_2} = \frac{16.2}{16.2 + 91} = 0.26016 \quad t = 600 \mu\text{m}$$

$$C_H = \frac{2(0.14)(7.77 \mu\text{m})(5.68)(0.26016)(0.0625)}{\left(600 \mu\text{m} - \frac{7.777}{1.56}\right)(6.5)} \times 10^6 = \boxed{51.953 \text{ Wt. Ppm}}$$

595.015

2. Limited by the diffusion of oxygen through the oxide layer to metal; "diffusion limited Process"

3. During operation hydrides preferably form circumferentially in the cladding and are concentrated at the rim due to soret effect (travel to low temps). This is due to free hydrogen formed by dissociation of water, from oxidation of cladding. The primary effect of hydrides is significant embrittlement of the cladding.

4. Reactivity insertion accident. Control Rod drop in BWR and Rod ejection in PWR. Immediate temperature increase which results in thermal expansion and pressure increase in gas bubbles, gap and clad. Additional fission rate provides more fission product swelling. When gap is more open in Low Burn up: DNB failure is likely. High burnup PCMI failure is more likely - would not be

5. Loss of Coolant Accident. During LOCA, the inability to cycle coolant leads to evaporation and cladding exposure to high temperature stagnant steam causing significant oxidation & hydrogen production. Additionally, loss of coolant leads to no outside pressure on cladding, so gap pressure creates outward force leading to ballooning & significant plastic deformation. LOCA is different than RTIA as it is a significantly slower acting process

6. Low to Mid-Low burnup fuel has significantly more enthalpy in the fuel and is of greater concern in RTIAs.

- Low Burnup fuel less likely to fail by PCMI due to less gap closure than high burnup fuel.

- High Burnup increased PCMI

- DNB may be more likely in Low-mid burnup due to less degraded fuel conductivity, but greater heat resistance due to gap size is a competing factor

7. 1. Improved oxidation kinetics (FeCrAl)

2. Improved cladding properties (Mechanical focus here)

3. Improved fuel properties (TiO₂ dopant)

4. Improved fission product retention

8. TiO_2 doping of UO_2 fuels (or other transition metals)

increases grain size of UO_2 improving thermal conductivity

(other is Cr/CrAl clad coating for oxidation resistance)

9. Rapid oxidation, oxidation process is exothermic and under

high temp steam breakaway oxidation can occur. As a result of this, clad temp is increased & embrittled by oxide layer and large hydrogen production which will accumulate for explosion risk and further embrittle the clad.

- wanted to sit more

10.

1. Cladding Oxidation/Wear. Must be less than 10% of

clad thickness and hydrogen production under 1% of total possible if all clad is oxidized

2. departure from nucleate boiling, DNB is often a limiting phenomena which expresses position in hottest channel's proximity to achieving DNB

3. PCMI, Pellet & Clad interaction is limited to prevent rupture. total hoop strain capped at 1%.

11. CRUD causes decreased thermal conductivity by adding another

layer of low conductivity between fuel and coolant. Additionally, CRUD components can get induced radioactivity, flake off and reach outside core components causing radiation safety risks.

Lastly, CRUD can trap neutron absorber Boron to clad causing crud induced power shift. CRIPs

12.

1. LiOH to increase pH to avoid CRUD formation (>6.9).
Also used in tandem with Boronic acid which is implemented for reactivity control but lowers pH. LiOH & Boronic acid concentrations are decreased simultaneously during operation to control pH.
2. Zn additions as it is more favorable in CRUD spinel (Al_2O_3) structure to replace cobalt or nickel which may become radioactive and leave the core. Zn is stable and allows dangerous metals to stay in pool for clean up.

13. 1. Central void. created by legumitic pores which travel up the thermal gradient and deposit in the center, ~~causing~~ causing large central void.

2. Columnar grain region outside central void caused by pores traveling up thermal gradient and "tearing" these grains into columnar shape

- this is two parts of same phenomenon

